UCLouvain

École polytechnique de Louvain

ÉCOLE POLYTECHNIQUE DE LOUVAIN

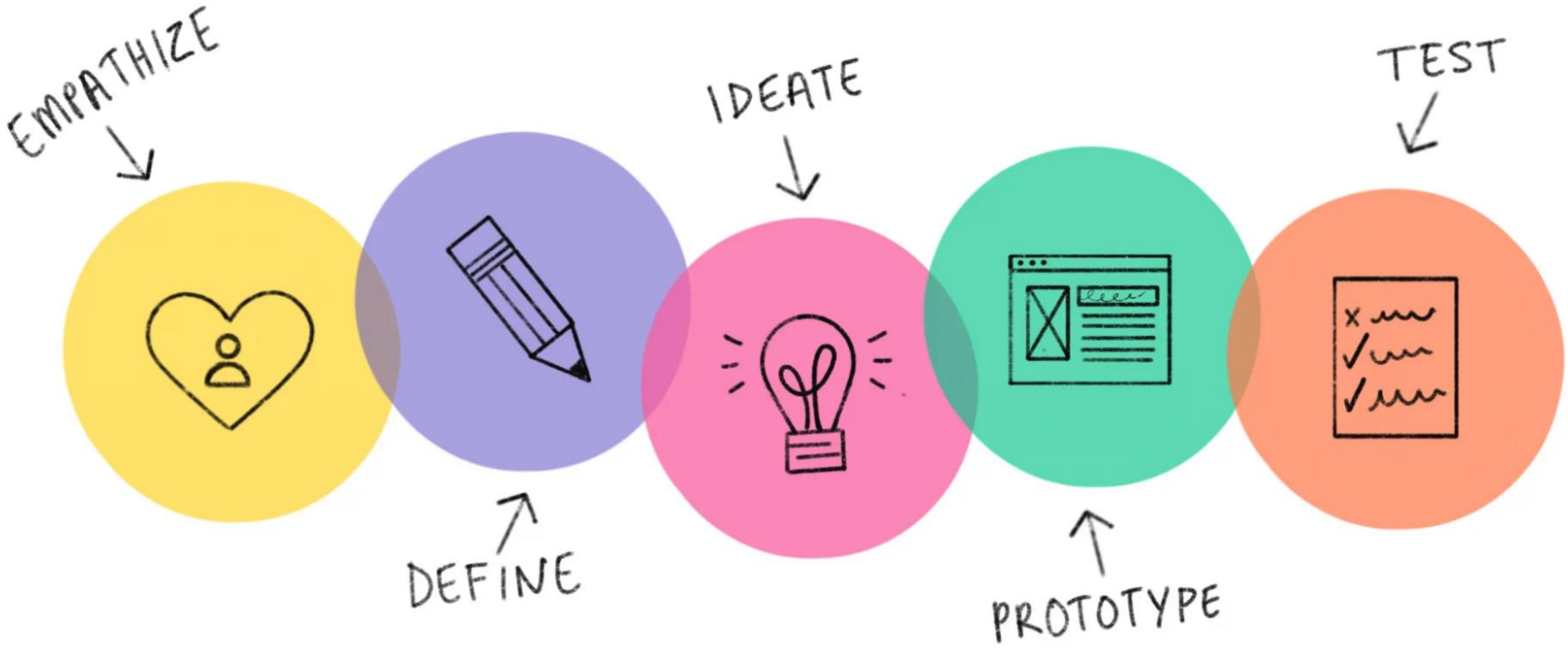
LEPL2022 - Health Innovation Classes Hackathon

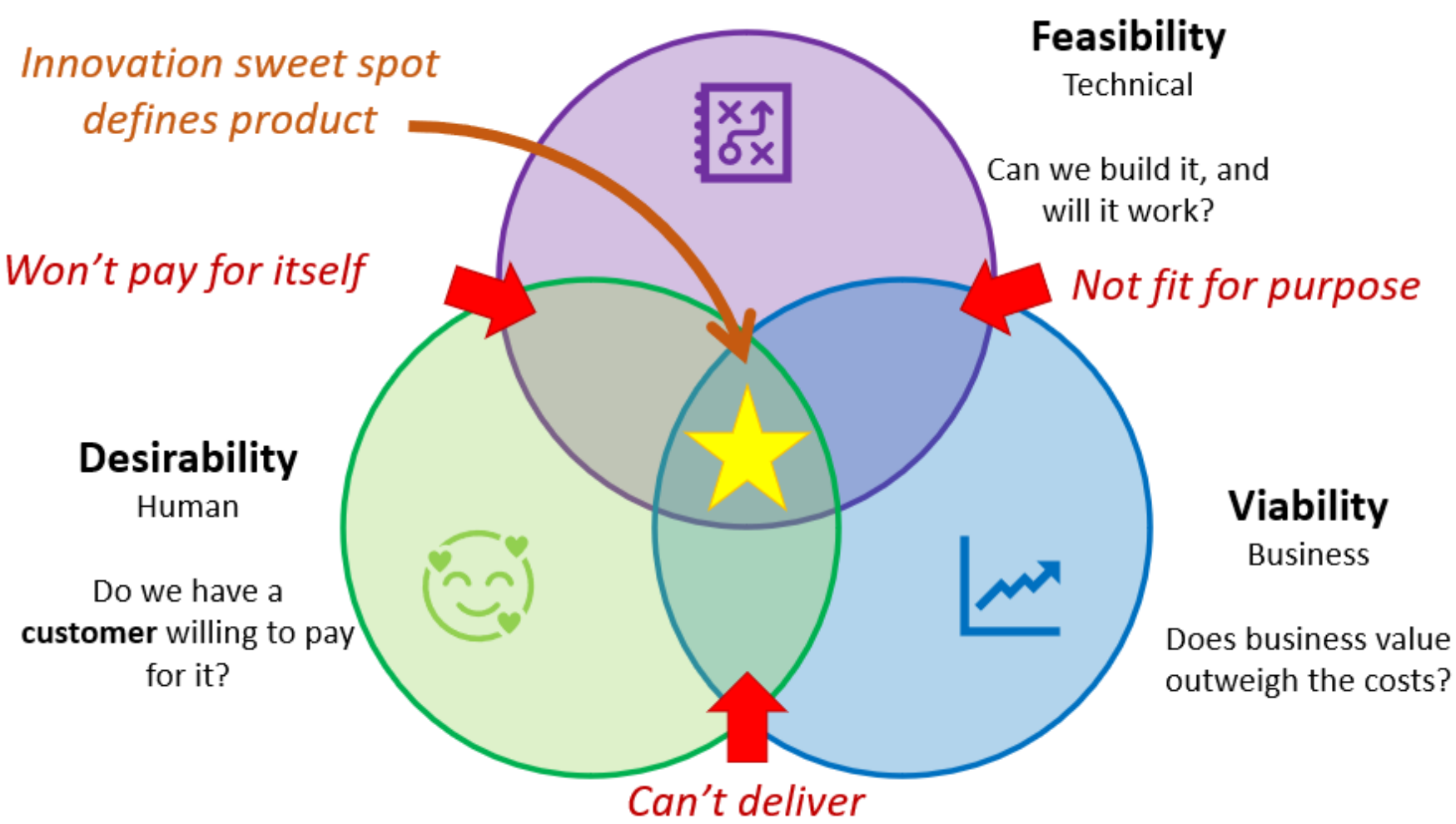
February 12th and 13th, 2026

Pierre Geens, MME, ieMBA

Reminder

DESIGN THINKING PROCESS




Innovation sweet spot defines product
Feasibility
Technical
Can we build it, and will it work?
Won't pay for itself
Not fit for purpose
Desirability
Human
Do we have a **customer** willing to pay for it?
Viability
Business
Does business value outweigh the costs?
Can't deliver

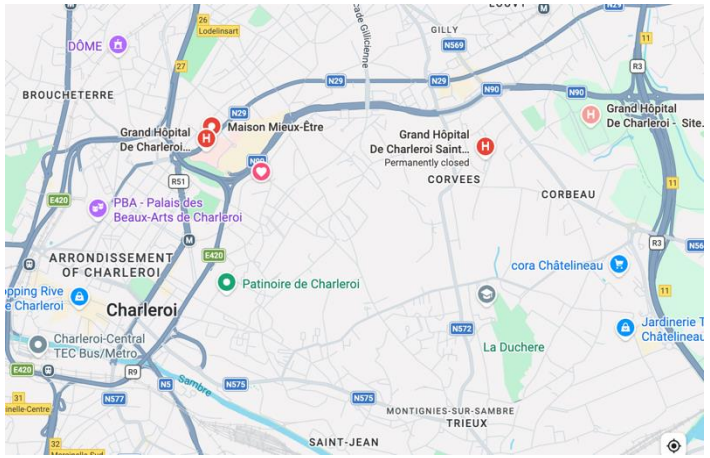
Hackathon

OBJECTIVES

- Go through the entire Design Thinking process in an accelerated manner
- Work on a concrete project. This year:

GHdC Central Pharmacy: Congestion and delays in handling returns of medicinal products

GRAND HOPITAL DE CHARLEROI



PLANNING

February 12th

- Arrival 8:30-8:45 at GHdC
- 9:00 start
- 9:00-9:45: Intro and presentation of the problem (incl. Field visit) + Q&A
- 9:45-10:00 Return to the room
- 10:00-12:00: Group work on empathy and problem definition
- 12:00-12:45: Lunch break (sandwich)
- 12:45-13:30: Another Q&A session
- 13:30-14:30: Iterate on empathy map and problem definition
- 14:30-16:00: Ideation and prototyping
- 16:00-17:00: Feedback from user representatives, presentation of the first concept of solution and prototype, feedback, Q&A
- 17:00-17:30 Wrap-up and closing of day 1, preparation for day 2

PLANNING

February 13th

- Arrival 8:30-8:45
- 9:00 Start
- 9:00-11:30: Work on the second iteration and associated prototype following feedback from the end of day 1
- 11:30-12:30: Final pitches
- 12:30: Lunch and departure

Reminder

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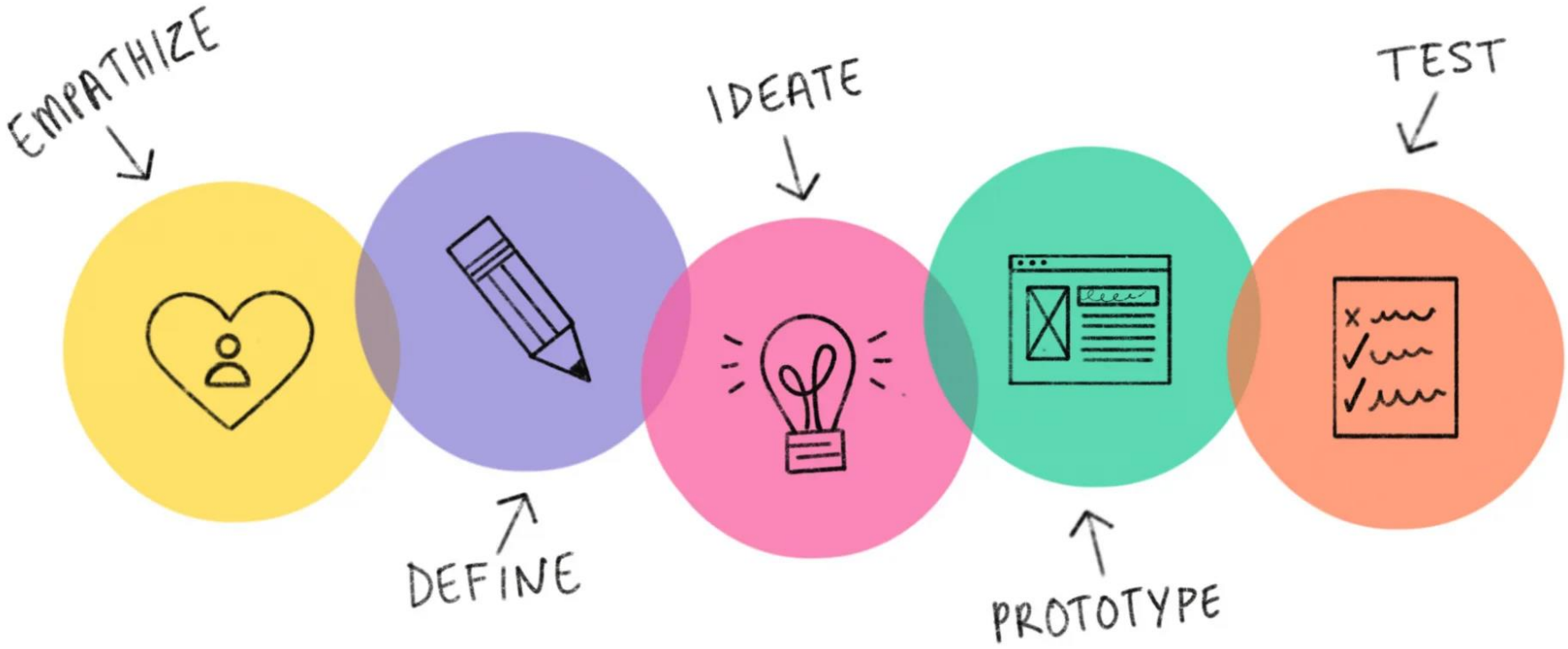

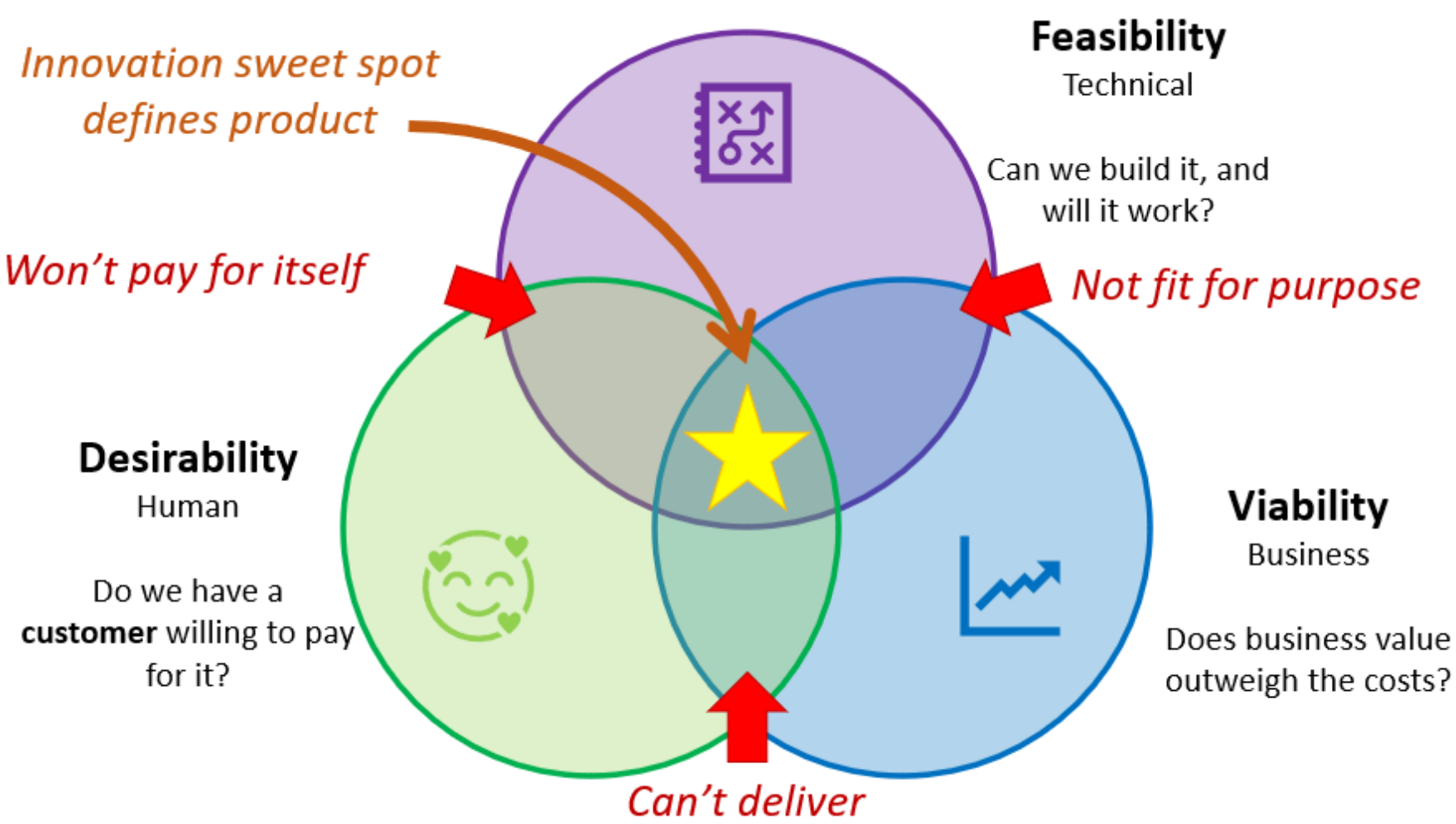
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Presentation of GHdC and of the challenge

Empathy and problem definition

Iteration #1

Empathize

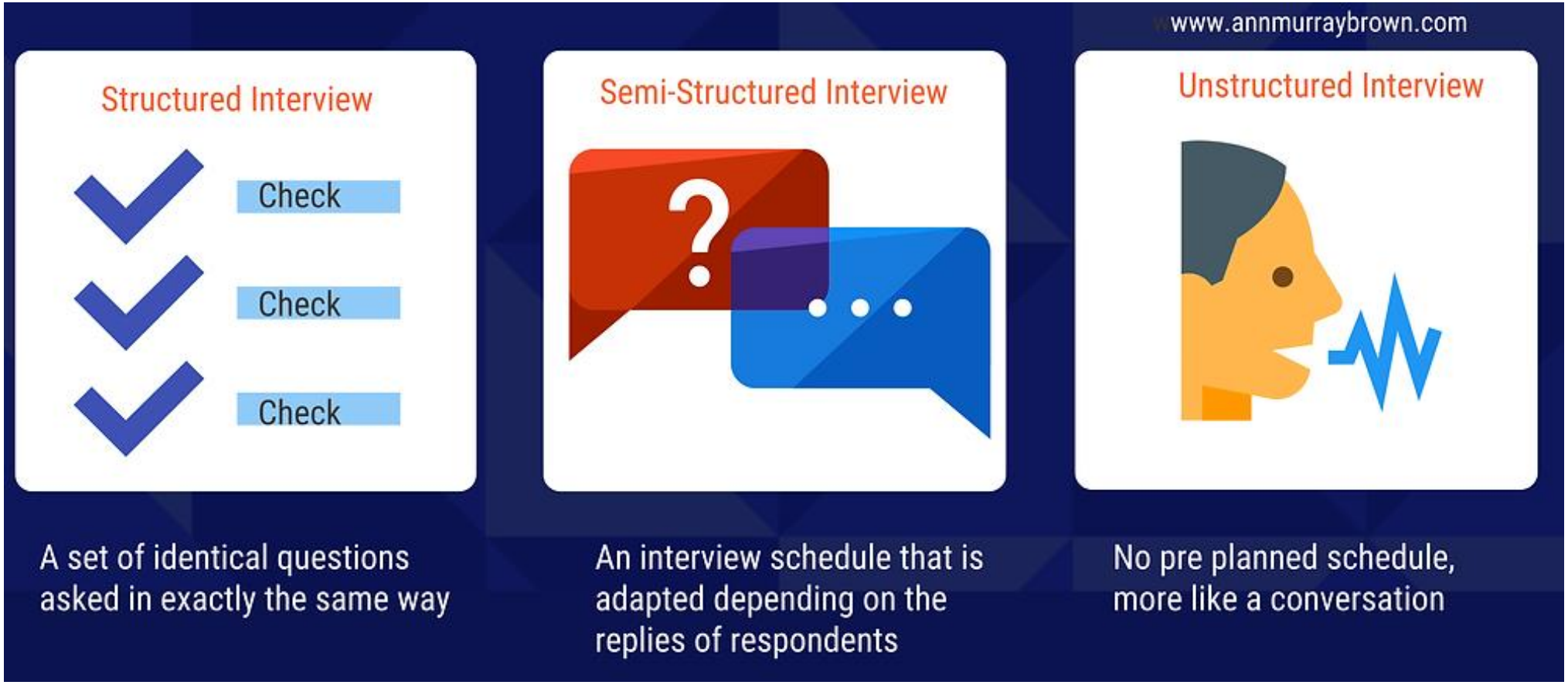
SEQUENCE

1. Identify stakeholders
2. Map stakeholders
3. Assess their respective influence and influence on the project
4. Select the most important
5. Create personas
6. Empathize
7. Create empathy maps

EMPATHIZE - SOURCES

- Primary sources
  - User interviews
  - Field observations / shadowing
  - Surveys
- Secondary sources
  - Scientific literature
  - Reports, statistics, benchmarks
  - Existing user feedback / complaints
- Contextual data
  - Environment, constraints, workflows
  - Social, cultural, institutional factors

EMPATHIZE – INTERVIEW METHODS



https://www.annmurraybrown.com/single-post/types-of-interviews

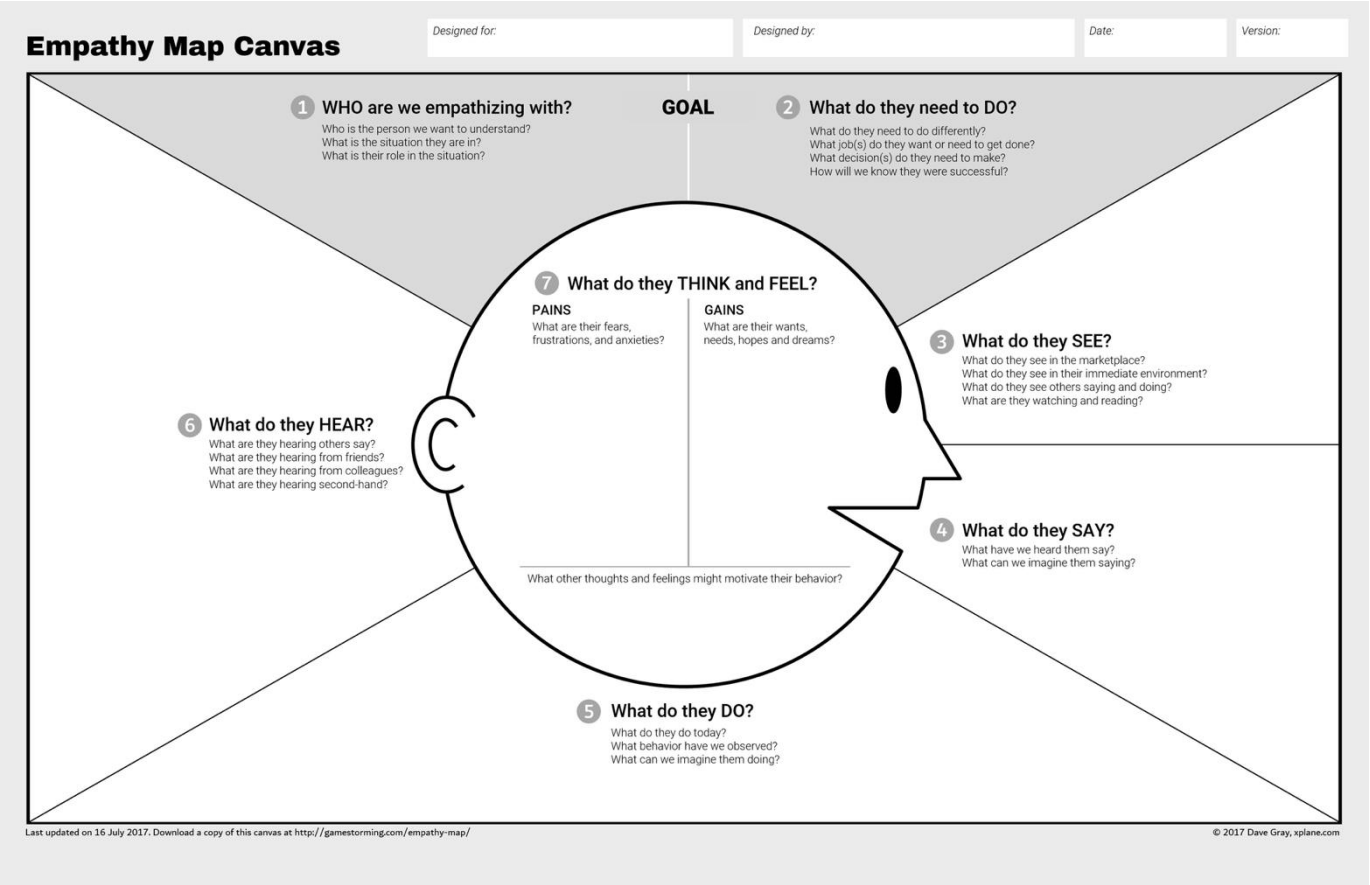
NON-STRUCTURED	SEMI-STRUCTURED	STRUCTURED
Characteristics • No fixed interview guide • Conversation-led, user-driven • High flexibility, high depth • Harder to compare or analyze systematically • Strong dependence on interviewer skill **Typical Questions** • "Tell me about a typical day." • "What does this situation mean for you?" • "Can you tell me more about that?" Mostly narrative prompts and follow-ups **Best Used In Design Thinking** **Stage:** Early Empathize **Situation:** • When the problem space is unclear • When exploring sensitive or complex contexts • When users are experts of their own experience	**Characteristics** • Interview guide with key themes • Flexible order and wording • Allows follow-up questions and probing • Balance between consistency and depth • Most common in design research **Typical Questions** • "Can you walk me through your last experience with…?" • "What was difficult or frustrating in that situation?" • "Why was that important to you?" Mix of open-ended and probing questions **Best Used In Design Thinking** **Stage:** Empathize / Define **Situation:** • Early problem exploration • When you want rich insights but still some structure • Small to medium sample sizes	**Characteristics** • Predefined questions, same order for all participants • Little to no deviation • Easy to compare answers across users • Lower interviewer bias, lower depth **Typical Questions** • "How often do you use this service?" • "On a scale from 1 to 5, how satisfied are you?" • "Which option do you use most frequently?" Closed or very specific questions **Best Used In Design Thinking** **Stage:** Test / Validate **Situation:** • When hypotheses are already defined • When you need comparability or metrics • When combining with quantitative analysis

EMPATHIZE – RESEARCH METHODS

QUANTITATIVE		QUALITATIVE
Numerical and measurable data	DATA TYPE	Descriptive and non-numerical data (words, images, observations)
Objective and structured measurement	APPROACH	Subjective understanding and interpretation
Large and representative	SAMPLE SIZE	Small, purposefully selected
• How many? • How much? • How often?	RESEARCH QUESTIONS	• Why? • How? • What does it mean?
Surveys, experiments, and closed-ended questionnaires	DATA COLLECTION	Interviews, case studies, open-ended surveys and observations
Statistical tools: averages, percentages, regression, etc.	DATA ANALYSIS	Thematic or narrative analysis, coding, grounded theory
Generalisable results and predictive insights	RESULTS	Deep, contextual understanding and rich detail

https://schoolofleadership.co.uk/quantitative-vs-qualitative-research

EMPATHY MAP



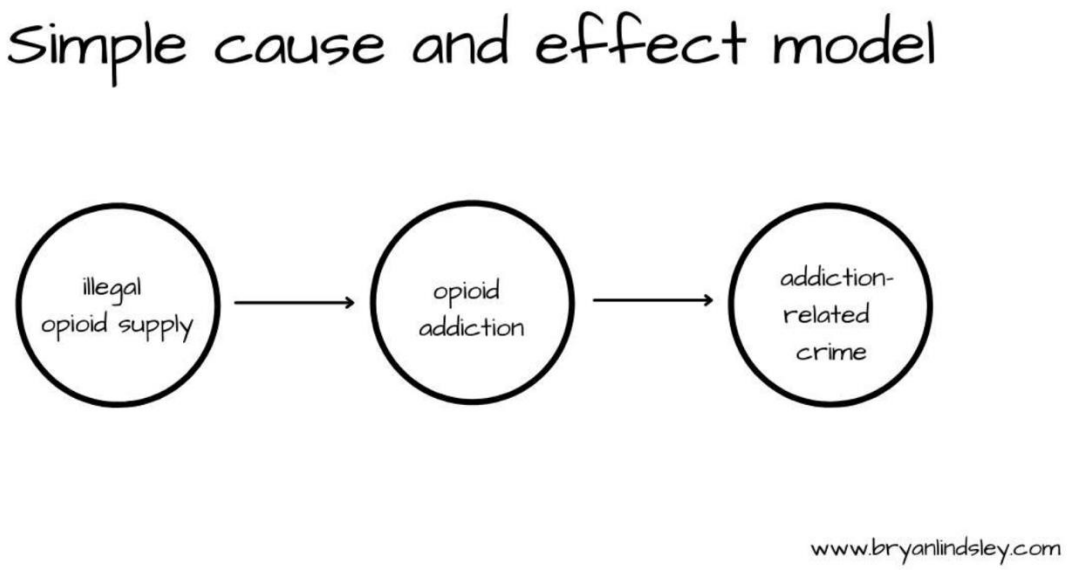
An empathy map helps teams understand users by capturing what they **think, feel, see, hear, say, and do**, revealing pains and gains to design solutions that truly fit their needs.

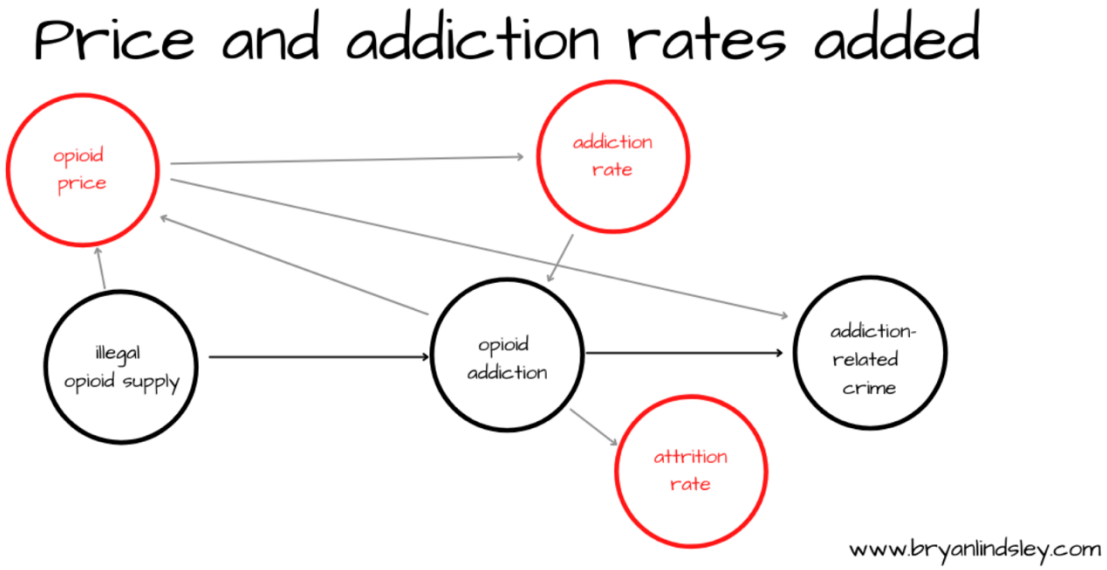
Problem definition

PROBLEM DEFINITION PRINCIPLES

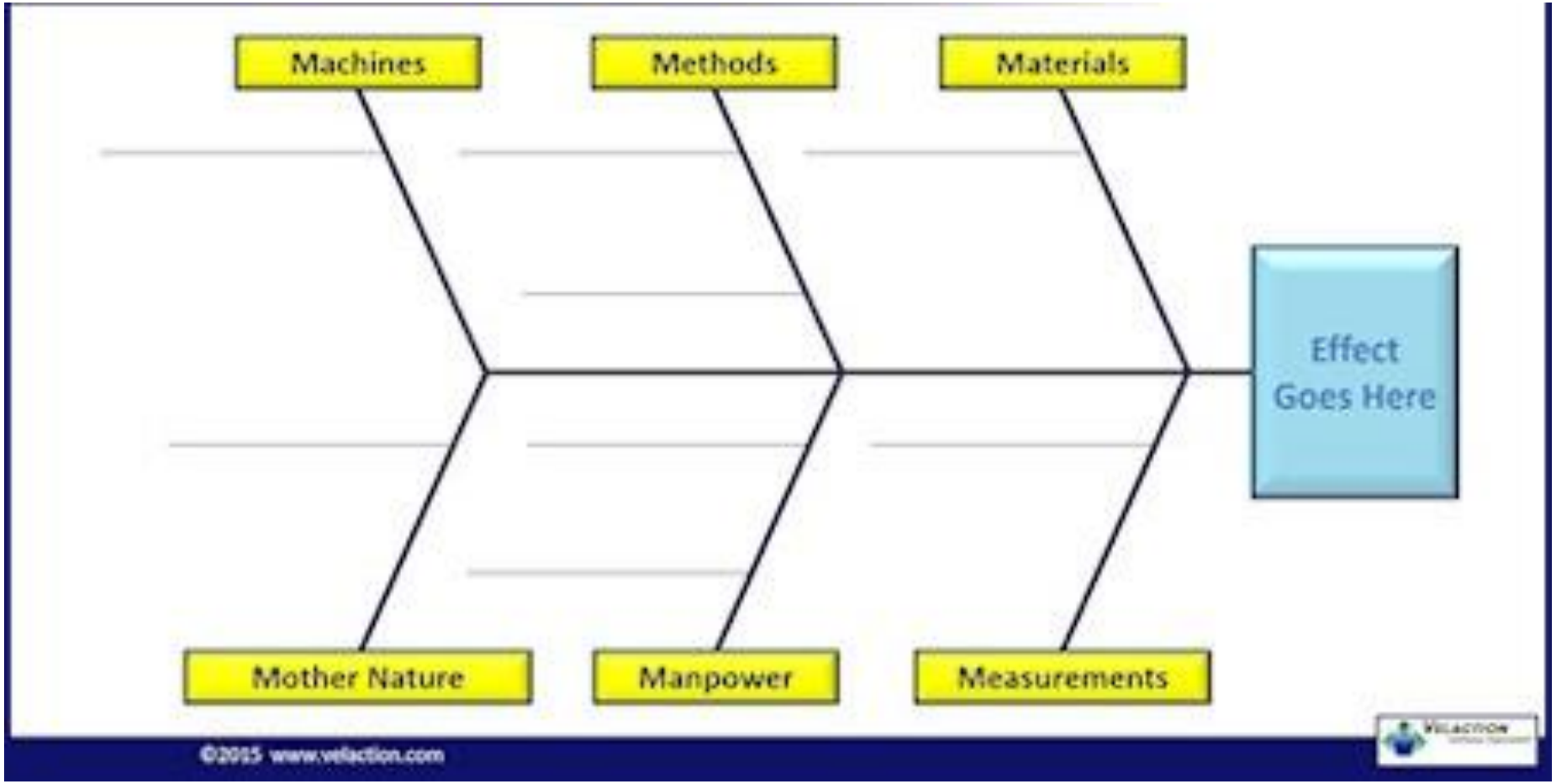
- **Human-centered,** focused on users' needs rather than solutions
- **Grounded in evidence** from research, observation, and interviews
- **Systemic**, accounting for interrelations and interdependencies between observations (actors, processes, constraints, context)
- Targets **root causes** and insights, not just visible symptoms
- **Clearly scoped**: neither too broad nor too narrow
- Framed as **a need**, not as a user failure or lack
- **Open-ended**, allowing multiple possible solutions
- **Shared and aligned** across the team and stakeholders

SYSTEMIC – Problem mapping





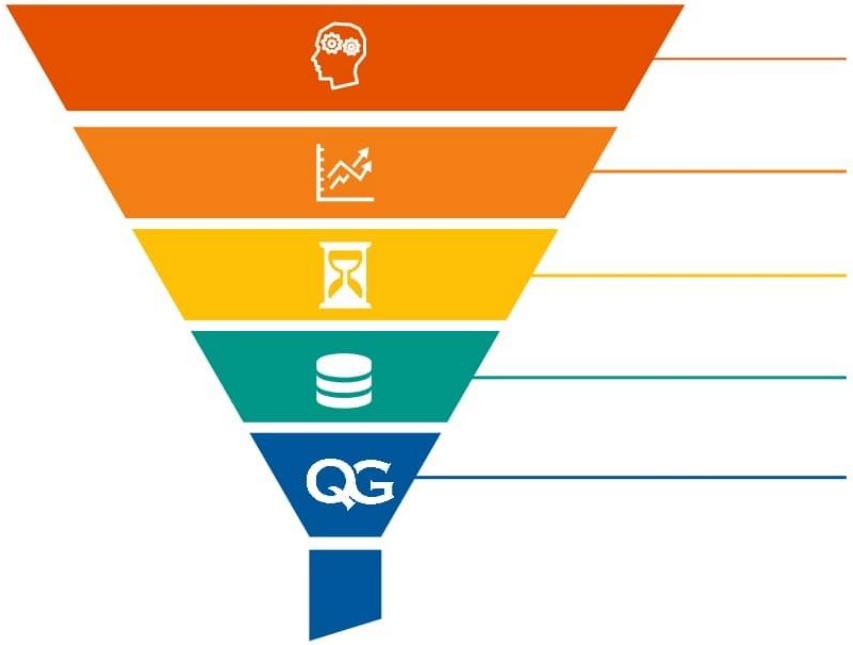
ROOT CAUSE – Fishbone diagram (Ishikawa)



ROOT CAUSE

5 Whys: Uncover Root Causes

The 5 Whys tool helps identify root causes of problems. However, misuse can lead to incorrect solutions. Learn how to avoid common mistakes.



1ST WHY
Identify the first cause contributing to the problem.

2ND WHY
Dig deeper by questioning why the first cause occurred.

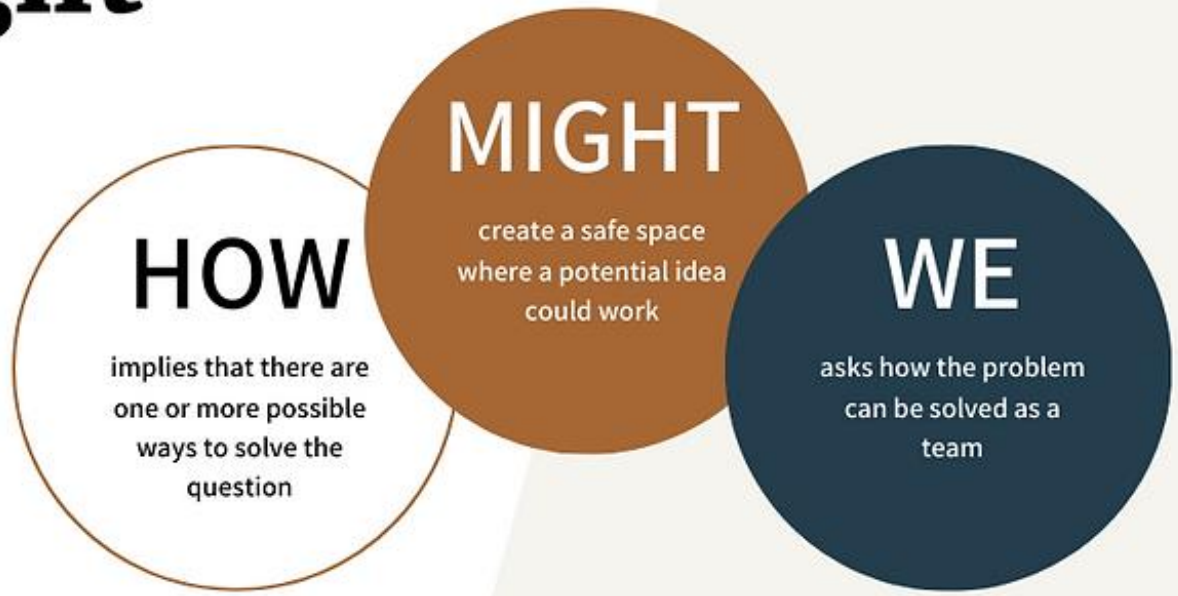
3RD WHY
Continue drilling down to understand deeper reasons.

4TH WHY
Reveal underlying system or process-related issues.

5TH WHY
Reach the actionable root cause and formulate corrective actions

PROBLEM STATEMENT

How Might We



PROJECT TAILWIND

PROBLEM STATEMENT – How might we

How might we _____
DO WHAT

for _____ in order to
FOR WHOM

_____ ?
BENEFIT, GAIN OR RESULT WE'D LIKE TO SEE

What you need to do

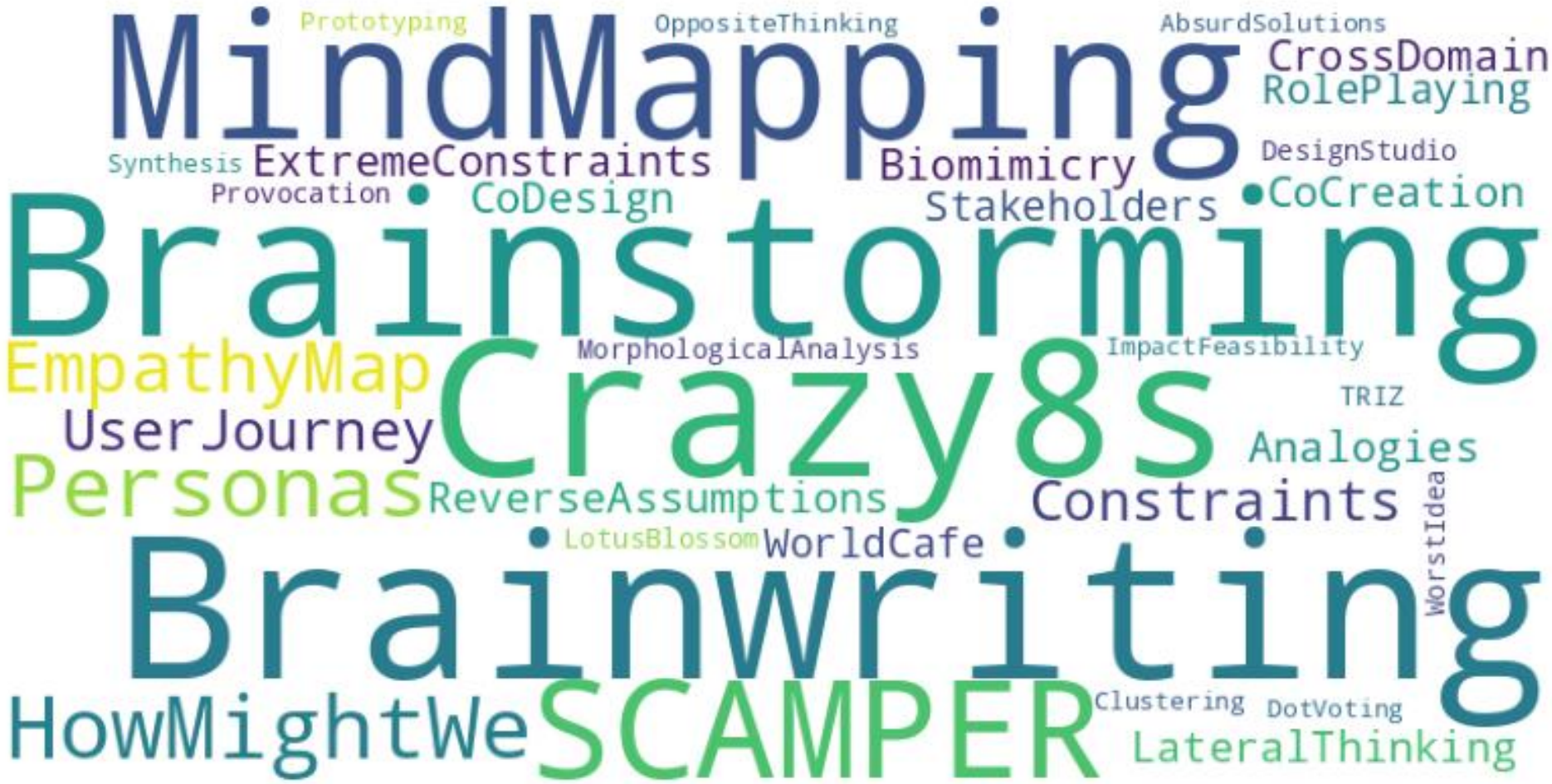
What you need to do – Iteration #1

- Gather all the insights from presentation and Q&A
- Build list of stakeholders
- Map stakeholders
- Conduct additional research (mostly secondary sources)
- First iteration of an empathy map
- Create a script for interview for the afternoon to enrich your empathy maps 10mins per group!!

Empathy and problem definition

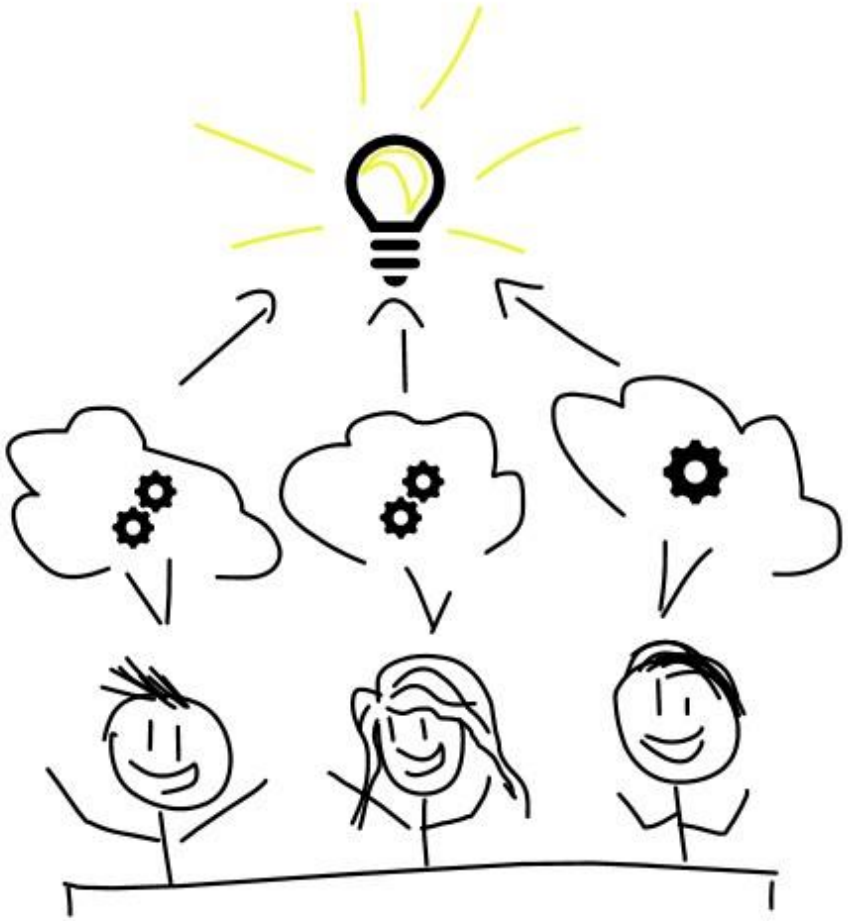
Iteration #2

Ideation


Prototyping
OppositeThinking
AbsurdSolutions
MindMapping
CrossDomain
RolePlaying
Synthesis
ExtremeConstraints
Biomimicry
DesignStudio
Provocation
CoDesign
Stakeholders
CoCreation
Brainstorming
EmpathyMap
MorphologicalAnalysis
ImpactFeasibility
Crazy8s
TRIZ
UserJourney
Analogies
Personas
ReverseAssumptions
Constraints
WorstIdea
LotusBlossom
WorldCafe
Brainwriting
HowMightWe
SCAMPER
Clustering
DotVoting
LateralThinking

BRAINSTORMING - DIVERGE

1. **Defer judgment** → no criticism
2. **Go for quantity** → more ideas = better chances
3. **Encourage wild ideas** → creativity first, feasibility later
4. **Build on others' ideas** → "Yes, and building on that…"
5. **Stay focused on the topic**
6. **Be visual** → sketch, post-it, draw



… THEN CONVERGE

1. **Cluster ideas** → group similar concepts (affinity mapping)
2. **Name the clusters** → reveal themes & directions
3. **Vote quickly** → dot voting, heat maps, silent voting
4. **Filter with criteria** → desirability, feasibility, viability ☺
5. **Select few ideas** → 1–3 concepts to prototype

DOT VOTING RULES

- **One dot = one vote**
- **Limited number of dots** per person (e.g. 3–5)
- **Vote silently**
- **You may split dots** or stack them on one idea
- **Time-box the vote** (1–2 minutes)

After voting:

- **Top ideas move forward**
- **No debate until voting is over**

Prototyping and testing

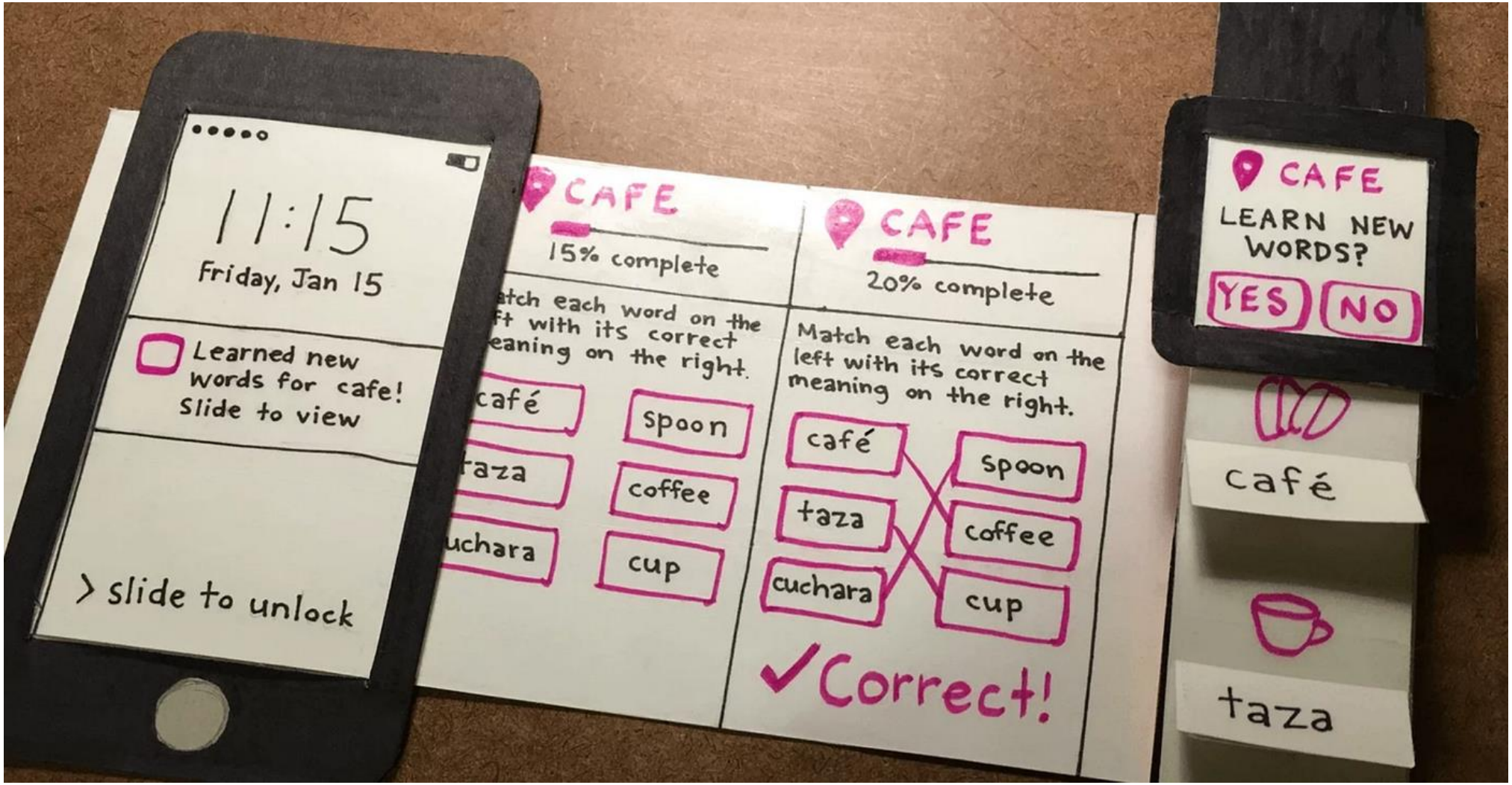
PROTOTYPING

Prototyping is about learning, not building the final solution!

- Make ideas tangible
- Test assumptions with users
- Get fast, concrete feedback
- Iterate and improve quickly
- Reduce risk early
- Align multidisciplinary team members

A GOOD PROTOTYPE IS

- **Fast & low-cost** → quick to build, easy to change
- **Disposable** → easy to discard or radically modify


11:15
Friday, Jan 15
Learned new words for cafe! Slide to view
> slide to unlock
CAFE
15% complete
tch each word on the ft with its correct eaning on the right.
café
taza
uchara
spoon
coffee
cup
CAFE
20% complete
Match each word on the left with its correct meaning on the right.
café
taza
cuchara
spoon
coffee
cup
✓Correct!
CAFE
LEARN NEW WORDS?
YES
NO
café
taza

A GOOD PROTOTYPE IS

- **Fast & low-cost** → quick to build, easy to change
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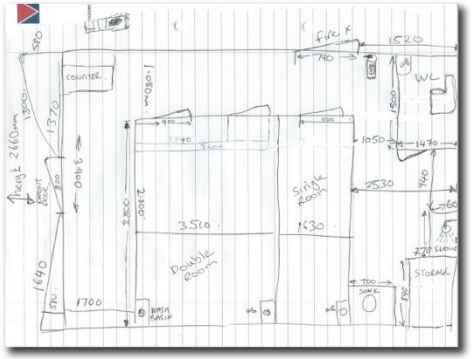
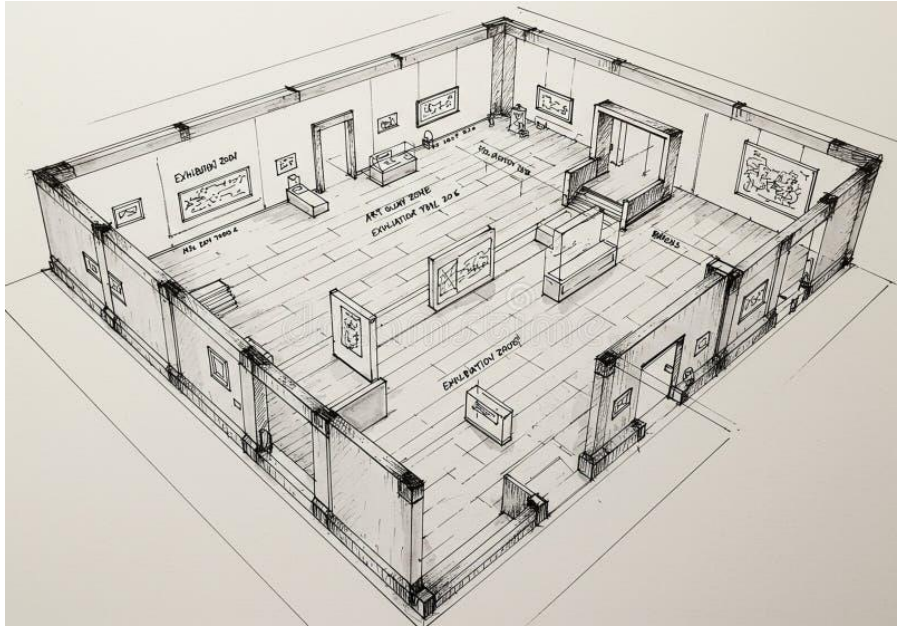
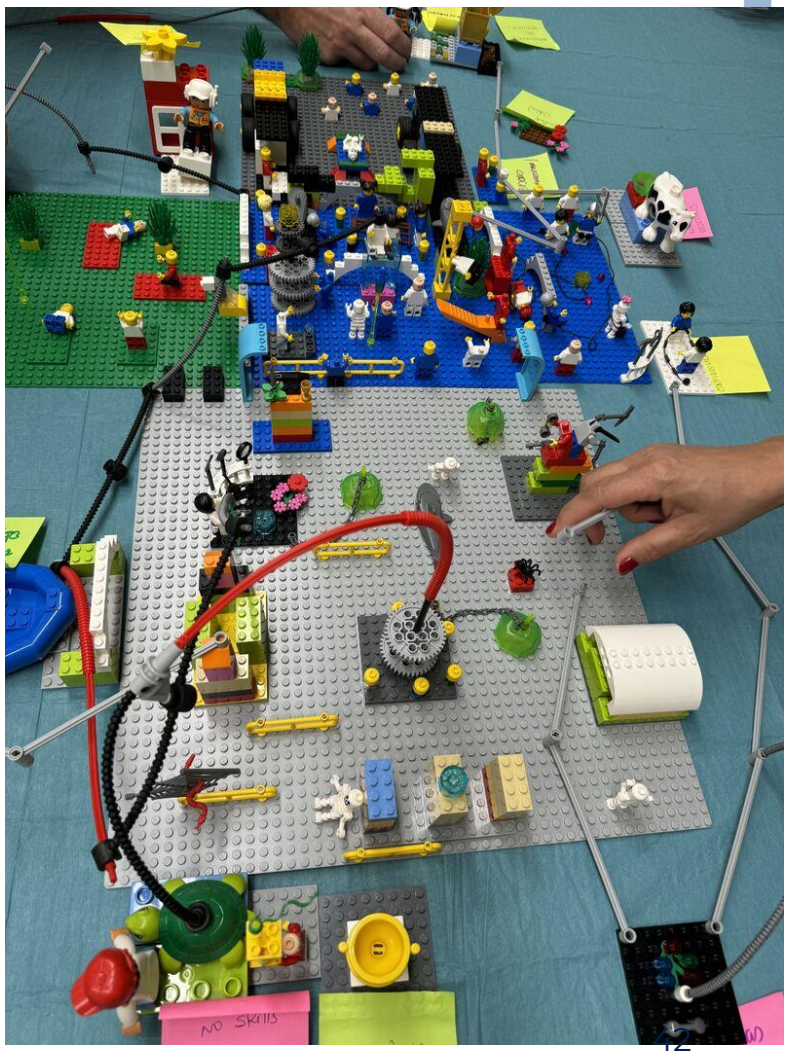
- **Understandable** → users grasp it without explanation
- **Testable with users** → enables observation & feedback


Fire X
1520
WC
Counter
Single room
Double room
Storage


Exhibition zone
Art gallery zone
Exhibition zone
Balcony


No skills

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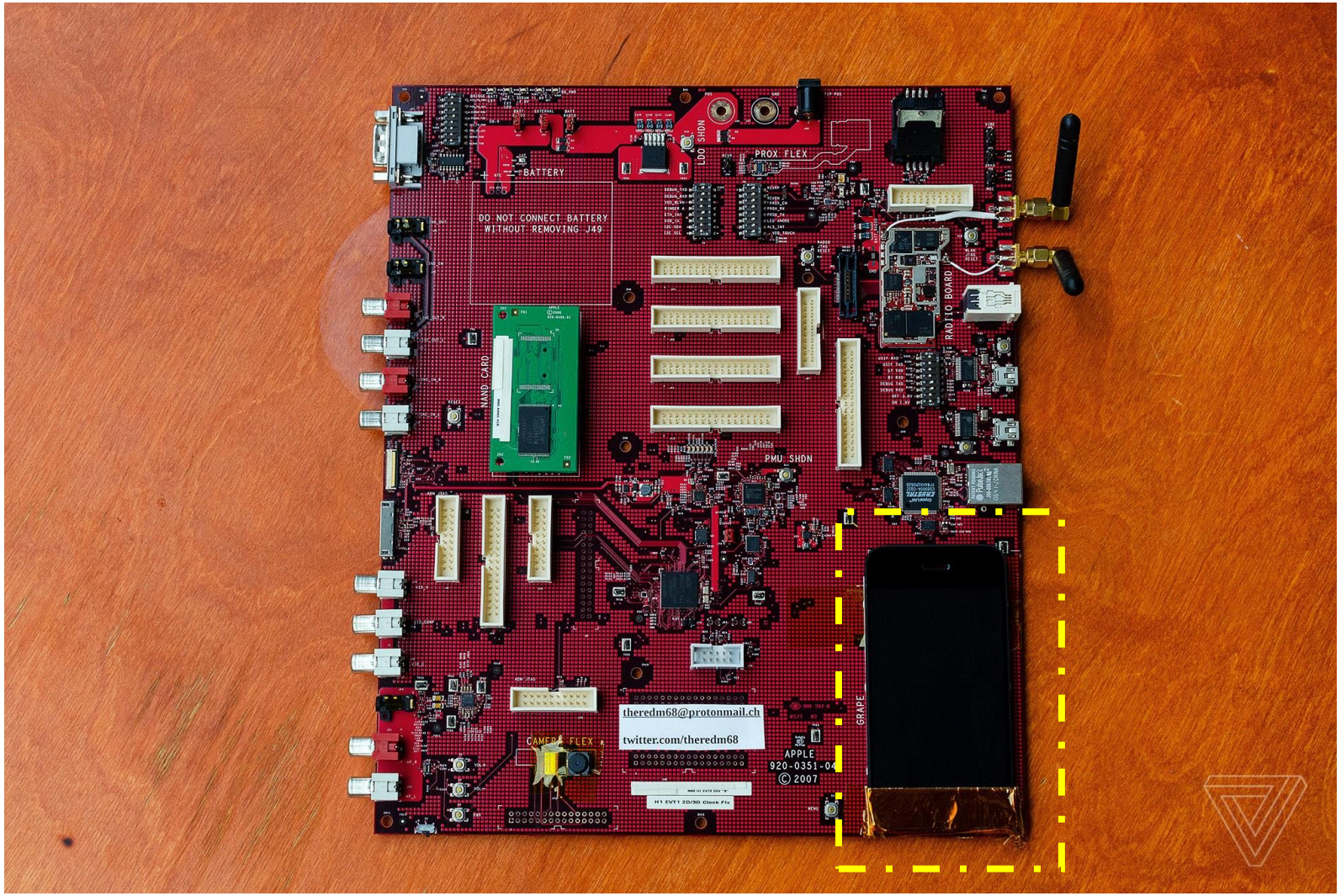
- **Purpose-driven** → answers a clear question
- **Focused** → tests one key assumption at a time


BATTERY
DO NOT CONNECT BATTERY
WITHOUT REMOVING J49
LDO SHDN
PROX FLEX
RADIIO BOARD
NAND CARD
PMU SHDN
theredm68@protonmail.ch
twitter.com/theredm68
CAMERA FLEX
GRAPE
APPLE
920-0351-04
© 2007

TESTING

Testing aims to:

- Observe real user behavior (not opinions)
- Validate or invalidate assumptions
- Gather concrete feedback
- Decide what to iterate, pivot, or stop

Key mindset:

- Test WITH users, not ON users
- The prototype is tested, not the person

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